Aim: To implement cryptography: symmetric encryption

What is cryptography?



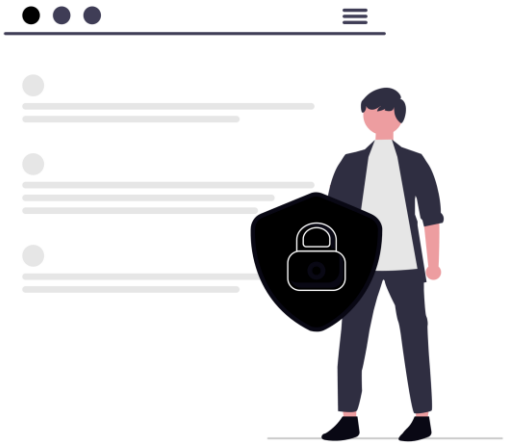
Cryptography is the scientific practice and study of techniques for securing communication and information against unauthorized access. It involves designing and analysing protocols that prevent third parties from reading private messages, verifying data integrity, and confirming identities.

Modern cryptography combines mathematics, computer science, and electrical engineering to protect everything from text messages to financial transactions. It forms the backbone of digital security, enabling safe online banking, e-commerce, and private communication across insecure networks like the internet.

What is meant by encryption?

Encryption is the specific process of transforming readable data, called **plaintext**, into an unreadable scrambled form, called **ciphertext**, using a mathematical algorithm and a secret value known as a **key**.

Define symmetric and asymmetric key encryptions.



In symmetric encryption, a **single shared key** is used for both encryption and decryption. The sender and receiver must both possess the same secret key beforehand.

Asymmetric encryption, also called **public-key cryptography**, uses a **pair of mathematically related keys**: a **public key** (shared openly) and a **private key** (kept secret). Data encrypted with the public key can only be decrypted with the corresponding private key, and vice versa.

How does a Ceaser Cipher work?

The **Caesar Cipher** is one of the oldest and simplest substitution ciphers, allegedly used by Julius Caesar to communicate with his generals. It works by **shifting each letter of the alphabet by a fixed number of positions**, called the **shift value** or **key**.

Only 25 possible keys exist, making it extremely easy to break via **brute force** (trying all shifts).

It is a **monoalphabetic substitution cipher**, meaning each plaintext letter always maps to the same ciphertext letter—vulnerable to **frequency analysis** (e.g., 'E' is the most common English letter, so the most frequent ciphertext letter likely represents 'E').

Code:

```
def caesar_cipher(input_text, key, encrypt=True):
    characters = ["A", "B", "C", "D", "E", "F", "G", "H", "I", "J", "K", "L", "M",
                  "N", "O", "P", "Q", "R", "S", "T", "U", "V", "W", "X", "Y", "Z",
                  "a", "b", "c", "d", "e", "f", "g", "h", "i", "j", "k", "l", "m",
                  "n", "o", "p", "q", "r", "s", "t", "u", "v", "w", "x", "y", "z",
                  "1", "2", "3", "4", "5", "6", "7", "8", "9", "0", "/", "+", " "]

    result = ""
    for char in input_text:
        if char in characters:
            if encrypt:
                result += characters[(characters.index(char) + key) % len(characters)]
            else:
                result += characters[(characters.index(char) - key) % len(characters)]
        else:
            result += char

    return result

pt = input("Enter a string: ")
key = int(input("Enter a key length: "))
encrypted = caesar_cipher(pt, key, True)
print("Encrypted:", encrypted)

ct = input("Enter cipher text: ")
decrypted = caesar_cipher(ct, key, False)
print("Decrypted:", decrypted)
```

Output:

```
spectre@ghouled-gadget MINGW64 D:/degree/SEM 7/SAD/Experiments/EXP10 (master)
$ py app.py
Enter a string: hello world
Enter a key length: 242466
Encrypted: xu225P 582t
Enter cipher text: xu225P 582t
Decrypted: hello world
```

```
spectre@ghouled-gadget MINGW64 D:/degree/SEM 7/SAD/Experiments/EXP10 (master)
$ py app.py
Enter a string: hello world
Enter a key length: 242466
Encrypted: xu225P 582t
Enter cipher text: lawdnla alksdna
Decrypted: VKgNXVKwKVUcNXK
```